



School of Computer Sciences Universiti Sains Malaysia

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CDS505 Data Visualisations & Visual Analytics

Project (25%)

Name	Matrix No.
Lee Chong Keat	23201641
Low Jun Hoe	23201640
Tee Li Hong	23201604
Teh Ming En	24102225

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1.0 Introduction and background

1.1 Link to Published Interactive Story Website

The interactive storytelling website developed for this project is published to the tableau public cloud and accessible via the following URL https://public.tableau.com/app/profile/chong.keat.lee/viz/USM_505_Project_final/Story1.

This web platform integrates the findings from the Labor Force Dashboard and the Crime Dashboard to present a comprehensive analysis of Malaysia's socioeconomic challenges from 2016 to 2023. The dashboard features interactive elements including state-level heat maps, trend lines, and district-level radar charts that allow users to explore the dynamic relationship between unemployment rates and crime incidents.

The period from 2016 to 2023 was a critical time for Malaysia, as the country faced unprecedented socioeconomic challenges. This era spanned political instability (Saidin, 2023), the global COVID-19 pandemic, the implementation of strict Movement Control Orders (MCO) (The Edge Malaysia, n.d.-b), and the subsequent economic recovery and inflation (Bank Negara Malaysia, n.d.). This report aims to provide a comprehensive analysis of the complex dynamic relationship between the Crime rate and the Labor Force during this 2016-2023 period.

1.2 Big Picture: Real-World Problem Context

Socioeconomic inequality and crime are deeply interconnected social issues that pose ongoing challenges for policymakers, researchers, and public institutions. Differences in environment, and labor force participation often vary significantly across regions and demographic groups, potentially influencing crime patterns and public safety outcomes.

Malaysia, with its diverse states and districts, presents a valuable case for exploring how economic conditions, labor force dynamics, and crime incidents interact spatially and temporally. While national statistics provide aggregate overviews, they often obscure regional disparities and localized patterns that are critical for effective policy intervention.

The complexity of this problem lies in the multivariate and temporal nature of the data. Understanding relationships between labor force participation, gender differences, and crime categories requires more than numerical summaries. Visual analytics and data storytelling allow human analysts to explore these relationships interactively, uncover hidden patterns, and communicate findings in a way that supports reasoning and decision-making.

This project applies visualization analysis and storytelling techniques to Malaysian socioeconomic and crime data to support tasks such as regional comparison, trend analysis, and pattern discovery. By transforming raw datasets into interactive visual narratives, the project aims to enhance understanding of socioeconomic disparities and their possible associations with crime across time and geography.

1.3 Dataset Description and Preparation

1.3.1 Dataset Description

The dataset used in this project is sourced from Kaggle (<https://www.kaggle.com/datasets/akma1xz/draft-dataset/data>) and compiles socioeconomic and crime-related data from Malaysia, covering the period 2016 to 2023.

The dataset consists of multiple related files:

1. **Crime incidents by district (`crime_district.csv`)** records the number of crime incidents in Malaysia at the state and district level, categorized by crime category and specific crime type from 2016 to 2023. This dataset enables fine-grained spatial analysis of crime distribution, as well as aggregation to the state level for regional comparison and temporal trend analysis across different crime categories.
2. **Labor force statistics by state and sex (`lfs_state_sex.csv`)** presents annual labor force statistics at the state level, disaggregated by sex. The attributes include labor force participation rates and unemployment rates for male and female populations from 1982 to 2022. This dataset supports analysis of gender-based labor market differences, regional employment patterns, and their evolution over time.

Together, these datasets enable cross-domain analysis between economic conditions, labor force dynamics, gender differences, and crime patterns at both state and district levels.

1.3.2 Dataset Preparation

To prepare the dataset for visual analytics and storytelling, a series of preprocessing steps were applied to ensure consistency, comparability, and analytical clarity.

1. Data cleaning was performed to address missing values, inconsistent formatting, and variations in state or district naming conventions across datasets. Records with incomplete essential fields were reviewed, and non-informative entries were excluded where appropriate to avoid misleading visual patterns.
2. Temporal and spatial alignment were carried out to enable meaningful cross-dataset analysis. All datasets were aligned by year, and common geographic identifiers such as state names were standardized to support joins between socioeconomic and crime data.

These steps ensure the data is suitable for interactive exploration without heavy programming.

1.4 Target Audience

The primary target audiences for this visualization project include:

1. Public policy makers and government agencies involved in socioeconomic planning and crime prevention.
2. Researchers and analysts in economics, criminology, and social sciences.
3. Data journalists and informed citizens interested in regional inequality and public safety trends.

The visualizations are designed for semi-expert users who understand the domain context but require visual support to interpret complex datasets.

2.0 Data Abstraction

Data abstraction is a critical phase in visualization design that involves translating raw data into a format suitable for visual encoding. Following the "What-Why-How" framework (Munzner, 2014), this chapter defines what data is available, classifying the dataset types, items, and attributes to guide the selection of appropriate visual channels.

2.1 Dataset Types

This project involves 3 dataset types:

1. Multivariate Tabular Data

The core structure of the dataset is multivariate and tabular, organized into rows (items) and columns (attributes). This format is essential for cross-domain analysis as it allows for the correlation of independent variables like socioeconomic factors with dependent variables of crime rates. The data is distributed across related files, such as `crime_district.csv` and `lfs_state_sex.csv`, which are linked through common keys.

For example, A single row in the labor force dataset represents a specific state-year observation (e.g., "Selangor, 2020"). This row contains multiple attribute columns such as the Labor Force Participation Rate (%), Unemployment Rate (%), and Total Employed Persons, allowing for a multidimensional assessment of that state's economic health in that specific year.

2. Temporal Data

The dataset is inherently temporal, spanning a longitudinal range from 2016 to 2023. This continuous timeline is critical for "change-over-time" analysis, enabling the detection of trends, cycles, and anomalies. The temporal granularity is annual, which supports the segmentation of analysis into distinct socioeconomic phases: the Pre-Pandemic era (2016-2019), the Pandemic/MCO era (2020-2021), and the Recovery era (2022-2023).

For example, the temporal dimension allows us to track the specific impact of the COVID-19 lockdowns (MCO). For instance, the data reveals a sharp 21.37% year-over-year decline in total crime cases in 2020, coinciding with a rise in the unemployment rate to 4.5%. Without this temporal component, such cause-and-effect relationships during crisis periods would remain invisible.

3. Spatial / Geographic Data

Although the dataset does not contain raw geometric coordinates (latitude/longitude), it utilizes nominal geographic identifiers specifically State and District names to establish a spatial structure. This data possesses a hierarchical nature, allowing for analysis at the granular District level and aggregation up to the State level. This supports spatial reasoning tasks, such as identifying regional hotspots and comparing urban versus rural disparities.

At the state level, the data distinguishes between highly urbanized regions like Selangor and Kuala Lumpur, which consistently record higher crime volumes, versus smaller regions like Perlis and Putrajaya. Furthermore, at the district level, the data allows for

drilling down into Petaling District, identifying it as a primary crime hotspot due to its high population density and complex road networks.

2.2 Data Items

Data items refer to the individual entities that are being visualized. Due to the integration of two different data sources, our items exist at two levels of granularity:

1. District-Level Items

In the crime dataset, an item represents a District-Year record like a record for Petaling District in 2019. These items contain fine-grained details about specific crime types (e.g., motorcycle theft counts).

2. State-Level Items

In the labor force dataset, an item represents a State-Year record like a record for Selangor in 2020. These items contain macroeconomic indicators such as labor force participation and unemployment rates.

3. Aggregated Regional Summaries

To enable cross-comparison between the two datasets, district-level crime items are often aggregated to match the state-level granularity. This results in derived items representing the total crime count for a state each year

2.3 Data Attribute Types

Attributes are the specific properties of the data items that are measured or recorded. Correctly identifying attribute types is essential for choosing the right visual channels, like using color hue for categories vs. color saturation for quantities.

1. Categorical Nominal Attributes

These attributes provide descriptive labels without an implicit order. They are primarily used for grouping and identification.

- a. Geographic Identifiers: State (e.g., Johor, Penang), District (e.g., Petaling, Kinta).
- b. Crime Category: A high-level classification dividing crimes into Property Crime and Assault Crime.
- c. Crime Type: Specific offenses such as Murder, Rape, Robbery, Vehicle Theft, and House Break-in.
- d. Demographic Group: Sex (Male/Female) used in labor force analysis.

2. Quantitative Ratio Attributes

These are numerical values where magnitude and ratios are meaningful. They are mapped to visual channels like position length of bars or color intensity heatmaps.

- a. Crime Counts: The absolute number of reported incidents like 34,754 motorcycle thefts.
- b. Unemployment Rate (%): A continuous value representing the percentage of the labor force without work.

- c. Labor Force Participation Rate (%): A measure of the active workforce relative to the population.
- d. Population/Labor Force Size: The count of individuals in thousands.

3. **Ordered Ordinal Attributes**

- a. Year: While numeric, Year 2016-2023 is treated as an ordinal attribute to define the sequence of the X-axis in trend lines and time-series charts.

2.4 **Derived Attributes**

Raw data alone is often insufficient for complex analytical tasks. To support deeper insights and reduce the cognitive load on the user, several attributes were mathematically derived from the original dataset.

1. **Year-over-Year (YoY) Change (%)**

To support Task 3 (Identify Relationships) and Task 4 (Compare Changes Over Time), we calculated the percentage change of crime and unemployment from the previous year. This derived attribute allows users to see the rate of change like a -21.37% drop in crime rather than just the absolute numbers, making trends during the pandemic immediately visible.

2. **Crime Rate Proportions**

We calculated the ratio of specific crime types to the total crime count. This supports Task 2, enabling users to understand the composition of crime in a state like what percentage of total crime in Selangor is Property Crime vs. Assault Crime.

2.5 **Comments**

The abstraction process highlights that this dataset is particularly robust for visual analytics. It is high-dimensional, combining economic, social, and criminal justice variables. It is geographically structured, allowing for spatial pattern discovery, and it is temporal, enabling the analysis of cause-and-effect relationships over time, such as the impact of COVID-19 lockdowns.

However, the primary challenge addressed during abstraction was the mismatch in granularity. Labor Force data exists only at the State level, while Crime data exists at the District level. The creation of Aggregated Regional Summaries (Section 2.2) was the key abstraction step that enabled the successful merging of these two domains for unified analysis.

3.0 Task Abstraction

Task abstraction defines what users need to do with the data in order to gain insight. Based on the crime, employment, and unemployment visualizations presented in our Tableau story, four high-level analytical tasks were identified. These tasks guide the design of the visual encodings and interactions used throughout the dashboard and storytelling sequence.

Task 1: Discover Regional Patterns and Trends

Task Type: Discover, Explore, Summarise

Charts Used:

- Choropleth map of total crimes by state
- Horizontal bar chart ranking states by total crime

The task enables users to identify geographical crime distribution patterns across Malaysian states and recognize which regions consistently experience higher or lower crime levels. Users explore a state-level map and bar charts showing total crime counts from 2017 to 2023. By interacting with the map (hovering, filtering by year or state), users can quickly discover regional crime hotspots such as Selangor and Kuala Lumpur, as well as states with comparatively lower crime incidence such as Perlis, Putrajaya, and Labuan. The visual aggregation allows users to move from a national overview to state-level insights, supporting pattern recognition across regions rather than isolated values.

Understanding regional crime patterns is essential for policymakers, law enforcement agencies, and researchers to prioritize resource allocation and intervention strategies.

Task 2: Compare Socioeconomic Conditions Across States and Demographic Groups

Task Type: Compare, Identify Differences

Charts Used:

- Employment rate by state (multi-year comparison)
- Unemployment rate by state
- Crime totals aligned with socioeconomic indicators

Task 2 is to compare employment rate, unemployment rate, and crime levels across different states to identify socioeconomic disparities. Users compare multiple states using grouped bar charts and tables that display employment and unemployment rates alongside crime totals. Filters allow users to isolate specific states or years, enabling cross-state comparison of socioeconomic indicators. This task highlights differences between highly urbanized regions and less developed states, revealing how employment conditions vary geographically and how these variations coexist with crime levels.

Socioeconomic inequality is often linked to social outcomes such as crime. This task helps

users understand structural differences between states rather than focusing solely on crime counts.

Task 3: Identify Relationships Between Socioeconomic Factors and Crime Patterns

Task Type: Identify Relationships, Correlate

Charts Used:

- Dual-axis line chart: Unemployment rate vs. total crime
- YoY percentage change in crime compared with unemployment trends

This task is to examine whether changes in unemployment rates are associated with changes in crime levels over time. Users analyze combined visualizations that overlay unemployment rates with total crime counts across multiple years. Line charts and dual-axis visuals allow users to observe whether increases or decreases in unemployment correspond to changes in crime.

Year-over-year (YoY) percentage change indicators further support this analysis by summarizing trends rather than requiring users to manually compare raw values. This task supports analytical reasoning potential socioeconomic drivers of crime, helping users form evidence-based interpretations rather than assumptions.

Task 4: Compare Changes Over Time

Task Type: Compare, Track Trends

Charts Used:

- Crime trends by year
- Crime type distribution over time
- Labor force and unemployment trends (2017–2021)

This task is to understand how crime and socioeconomic indicators evolve over time, particularly before, during, and after significant periods such as the COVID-19 years. Users use time-series visualizations to track crime counts, crime categories (property vs. assault), labor force participation, and unemployment rates from 2017 to 2023. Interactive year filters and sliders allow users to focus on specific periods and observe shifts in trends.

This temporal comparison reveals notable declines in crime during certain years and gradual recovery or changes afterward. Temporal analysis helps users distinguish short-term fluctuations from long-term trends, supporting more accurate interpretation of policy or economic impacts.

4.0 Design Justification

4.1 Marks and Channels

Following Munzner's visual encoding framework, the visualisations mainly use basic geometric marks, including bars, lines, points, and geographic areas. These marks were paired with common visual channels such as position, length, color hue, color intensity, and spatial location. For quantitative attributes like crime counts and unemployment rates, positions on a common scale and length were used as the primary channels, as these channels support more accurate visual judgement. Color hue was mainly used to distinguish categorical variables such as states or crime types, rather than to encode precise numerical values. This helps reduce misinterpretation and keeps the overall design visually consistent.

4.2 Visual encoding and analytics for each task

4.2.1 Task 1: Discover Regional Crime Patterns and Trends

For this task, bar charts and a map were selected to support pattern discovery across regions.

Marks	Bars and geographic map regions
Channels	Position (X-axis): State Position (Y-axis): Total crime count Color hue: State or crime category Geographic position: Geographic location of states

The bar chart allows users to compare crime levels across states using bar length, making it easy to identify high-crime regions such as Selangor and W.P. Kuala Lumpur. The accompanying choropleth map reinforces spatial patterns by encoding crime intensity using color saturation. This combination supports pattern discovery by allowing users to move between numerical comparison and spatial reasoning without changing context.

4.2.2 Task 2: Compare Crime Types Across States

To compare different crime categories within and across states, stacked bar charts were used.

Marks	Stacked bars
Channels	Position (X-axis): State Position (Y-axis): Total crime count Bar length: Total crime volume Color hue: Crime type

Stacked bars were chosen to support part-to-whole comparison, enabling users to see both the overall crime level of a state and how it is composed of different crime types. Color hue provides categorical separation between crime types, while the total bar length still allows comparison across states.

4.2.3 Task 3: Identify Relationships Between Socioeconomic Factors and Crime

This task is to explore the relationship between unemployment rate and crime over time, a dual-axis line chart was used.

Marks	Lines and points
Channels	Position (X-axis): Year Position (Y-axis): Total crime counts and Unemployment rate Color hue: Measure type (crime vs unemployment)

The line chart was used to visually align unemployment trends with crime trends over time. By aligning both measures on the same time axis, users can visually compare their movements across years. Year-over-year (YoY) percentage change indicators were added to summarise directional change and help users focus on overall patterns rather than individual data points. This design supports relationship identification and trend reasoning rather than causal claims.

4.2.4 Task 4: Compare Changes Over Time

For analysing how crime levels change across multiple years, line charts were again used.

Marks	Lines
Channels	Position (X-axis): Year Position (Y-axis): Crime count Color hue: Crime category or state Position: Temporal progression

Line charts are well suited for change-over-time analysis, by using vertical position makes increases and decreases easy to detect. Consistent color encoding across years ensures that users can track the same category without confusion. This design supports temporal comparison and trend identification, which are central to this task.

4.3 Interaction Design and Analytical Support

Interactivity was added to support visual analytics and user exploration. Filters allow users to select specific states, years, or crime types, which helps narrow down the analysis. Hover tooltips provide exact values when needed, reducing visual clutter. Linked views ensure that selections made in one chart are reflected in others, helping users maintain analytical context. These interaction techniques follow Munzner's concept of supporting overview first, zoom and filter, then details-on-demand.

In summary, the visual design follows Munzner's principles of using appropriate marks and perceptually effective channels to support analytical tasks. Position and length were prioritised for quantitative comparison, while color was used mainly for categorical distinction. Combined with interactive features, these design choices enable users to explore patterns, compare categories, and analyse trends effectively.

5.0 Four Levels of Validation

To ensure that the interactive visualization and storytelling system is credible, accurate, and analytically meaningful, this project adopts a four-level validation framework grounded in established visualization research. The validation process addresses domain relevance, data and task correctness, visual design effectiveness, and algorithmic reliability, aligning with best practices in visualization analysis and design (Munzner, 2014). By following this structured approach, the project ensures that the resulting insights are not only visually compelling but also methodologically sound and actionable for decision-makers.

5.1 Domain Validation

The objective is to ensure that the problem framing, analytical focus, and interpretations are meaningful and relevant to the real-world domain of environment that focus on socioeconomic development and crime analysis in Malaysia.

The project is grounded in well-established domain concerns, specifically the interaction between crime patterns, unemployment, labor force participation, and regional socioeconomic inequality. Indicators selected for visualization such as total crime counts, crime categories (property and assault), employment rates, unemployment rates, and labor force size are widely used in public policy and social science analysis.

Domain validity was further reinforced by evaluating whether patterns observed in the final dashboards align with known characteristics of Malaysia's socioeconomic landscape. For instance, the visualizations consistently show higher crime volumes in highly urbanized and densely populated regions such as Selangor and Kuala Lumpur. In contrast, smaller administrative regions such as Perlis, Putrajaya, and Labuan exhibit substantially lower crime levels, which is consistent with expectations given their population size and urban scale.

Similarly, the unemployment and labor force dashboards reveal increased unemployment rates during the COVID-19 period (2020–2021) followed by stabilization in subsequent years. These patterns align with publicly reported economic impacts during that period, increasing confidence that the visual narratives are domain-plausible. Importantly, the visualizations are framed to support exploration and hypothesis generation, avoiding unsupported causal claims between socioeconomic conditions and crime.

The visualization system reflects realistic domain behavior and supports meaningful interpretation by policymakers, researchers, and informed citizens.

5.2 Data and Task Validation

The objective is to ensure that the datasets used are appropriate for the intended analyses and that the defined analytical tasks can be effectively performed using the available data and visualizations.

The second level of validation ensures that the selected datasets and abstracted tasks are logically aligned to address the research objectives. The suitability of the datasets was rigorously assessed based on their capacity to support three essential analytical dimensions: spatial,

temporal, and multivariate. Specifically, the data provides granular state and district-level identifiers for spatial analysis, a longitudinal scope spanning 2016 to 2023 for temporal trend detection, and a rich set of variables which includes crime rates and labour force indicators to facilitate multivariate comparison. By harmonizing these disparate data sources, the project creates a robust multidimensional framework capable of capturing the complex socioeconomic dynamics of Malaysia (Munzner, 2014).

The preprocessing phase was critical to ensuring the validity of this abstraction. This workflow involved data cleaning, the standardization of geographic nomenclature, and temporal alignment to resolve inconsistencies between different reporting agencies. A key methodological step was the systematic aggregation of district-level crime data to the state level. This is important to achieve parity with socioeconomic indicators which were only available at the state tier. Furthermore, normalization techniques were applied to ensure that comparisons across regions with varying population densities remained statistically valid, following the "Tidy Data" principles of ensuring each variable forms a column and each observation forms a row (Wickham, 2014).

To bridge the gap between domain questions and visual design, task abstraction was validated by mapping real-world analytical needs to specific visualization operations.

- **Discovery of regional patterns:** Facilitated through choropleth maps and state-ranking bar charts.
- **Socioeconomic benchmarking:** Enabled via employment heatmaps and comparative bar charts for cross-regional analysis.
- **Relational identification:** Exploring the nexus between unemployment and crime through dual-axis line charts and Year-over-Year (YoY) visualizations.
- **Temporal tracking:** Supported by time-series charts to visualize fluctuations in crime types and labour force trends.
- **Intra-state crime composition analysis:** Supported by the district-level crime distribution (sunburst) visualization that allow users to examine how individual districts contribute to total state crime.

This explicit mapping ensures that every interactive component in the Tableau story serves a verified analytical purpose, allowing users to transition from high-level overviews to granular insights without compromising data integrity.

5.3 Design Validation

The objective is to ensure that visual encodings, layouts, and interactions effectively support user tasks and reduce cognitive load.

The third level of the nested model evaluates the effectiveness of the chosen visual idioms and interaction techniques. Visual encodings were selected based on the principles of graphical perception, ensuring that the most important data attributes were mapped to the most effective visual channels. For instance, bar charts were prioritized for comparing crime volumes across states because length is a highly accurate channel for quantitative comparison. Line charts were utilized to emphasize temporal trends and Year-over-Year (YoY) shifts, leveraging the human eye's ability to perceive slope and directionality in time-series data (Munzner, 2014). Furthermore, heatmaps and choropleth maps were deployed to provide immediate spatial context, enabling users to rapidly identify regional disparities and outliers in employment and crime distribution without excessive cognitive effort. A hierarchical radial chart (sunburst) was employed to represent crime distribution by district within each state. This design choice supports part-to-whole

reasoning, allowing users to understand how individual districts contribute to overall state crime while maintaining contextual awareness of higher-level aggregation. The concentric layout enables simultaneous comparison across districts without requiring users to navigate away from the state-level perspective.

To ensure a cohesive user experience, the system adheres to the expressiveness and effectiveness criteria. Design consistency was maintained through standardized colour palettes for specific crime categories and socioeconomic indicators, coupled with clear axis labelling and uniform dashboard layouts. These choices minimize the gulf of evaluation that allows users to interpret the visualizations intuitively. Interaction design was validated by providing an initial overview of the Malaysian landscape, followed by filtering and zooming capabilities such as state selectors, and finally offering the details-on-demand via informative tooltips. These interactive features allow for progressive exploration, ensuring that the complexity of multivariate data does not overwhelm the user.

The final stage of design validation involved an informal usability walkthrough conducted from a first-time user perspective. This process confirmed that critical insights such as the concentration of crime in urban centres like Selangor or the economic shocks observed during the COVID-19 period were discoverable without external guidance. The outcome of this level of validation is a design that successfully bridges the gap between complex raw data and the mental models of semi-expert users, facilitating both trend analysis and pattern discovery.

5.4 Algorithm Validation

The objective is to ensure that data transformations, aggregations, and derived metrics are computed correctly and do not introduce analytical bias.

The final level of validation focused on the computational integrity of the data pipeline to ensure that all underlying transformations, aggregations, and derived metrics were executed with mathematical precision. The primary concern was the reconciliation of spatial granularity. Since socioeconomic indicators were provided at the state level while crime data was recorded at the district level, systematic aggregation logic was implemented. To verify this, summed district-level incidents were cross-referenced against official raw data summaries to ensure no data loss or duplication occurred during the integration phase. Furthermore, normalization techniques were validated to account for population disparities across Malaysian states to ensure that the resulting crime rates offered a fair and equitable basis for regional comparison.

To ensure the reliability of the visual narratives, the accuracy of high-level Key Performance Indicators (KPIs) was subjected to rigorous manual verification. Core metrics including Total Crimes, Property Crimes, and Assault Crimes were checked for specific states and years to confirm that the values rendered in the Tableau dashboards mirrored the source datasets. This verification extended to complex derived attributes, such as Year-over-Year (YoY) percentage changes in unemployment and crime. By manually recalculating these growth rates, the study confirmed that the trends depicted in the dual-axis and comparative visualizations were grounded in verified arithmetic, thereby mitigating the risk of misleading visual interpretations.

Finally, the functional stability of the interactive system was evaluated by testing the behavior of derived attributes under varying filter conditions. This involved stress-testing the dashboards with multiple concurrent selections such as specific year ranges, state filters, and crime categories to

ensure that the summaries for labor force trends and crime composition remained consistent and computationally stable.

This level of verification ensures that the interactive storytelling system is not merely a visual artifact but a reliable analytical tool, where every data point is accurate and every calculation is robust enough to support evidence-based reasoning.

6.0 Final Visualisations

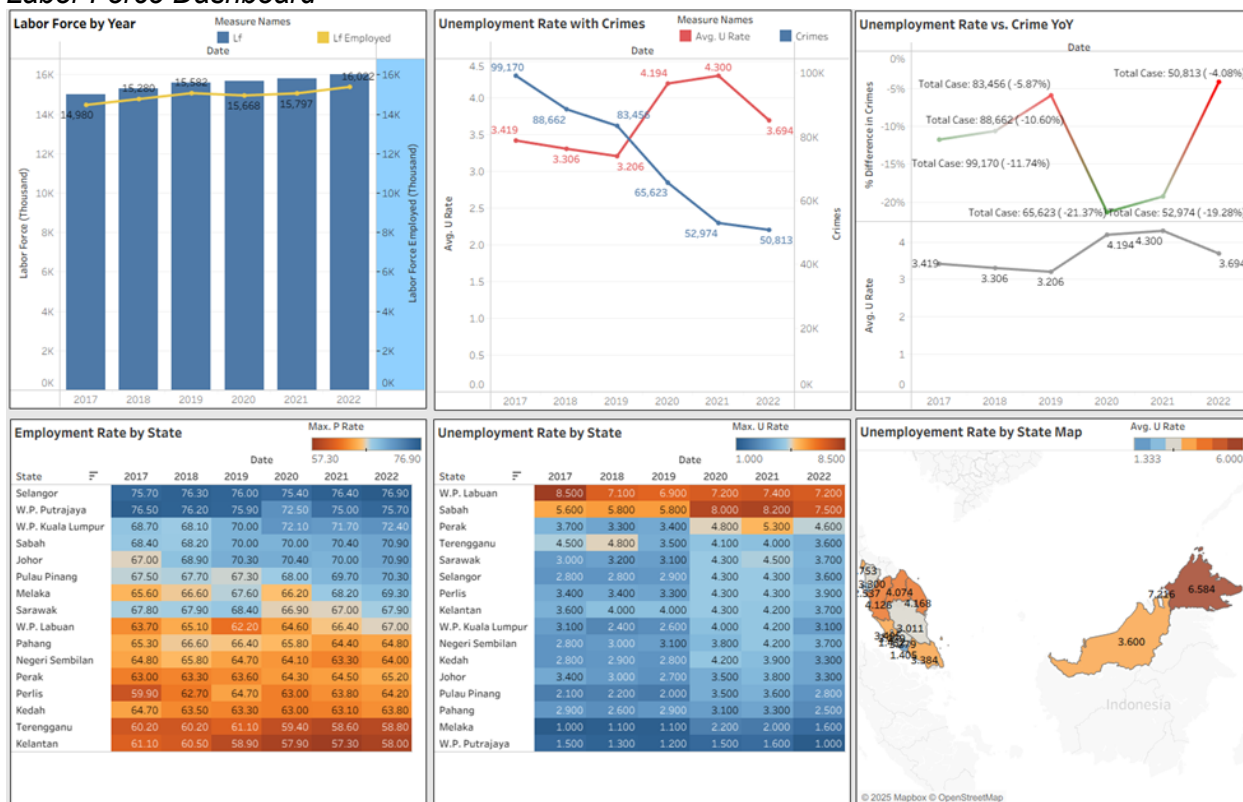
Traditional criminological theories, specifically Merton's Strain Theory and Becker's Economic Theory of Crime, typically assume a strong positive link between unemployment rates and crime rates. The core logic is because without a job or salary, people have less to lose, so the opportunity cost of turning to crime is lower.

However, based on our data analysis, this study found that this classic theory failed during the crisis in Malaysia from 2020 to 2021. While the unemployment rate climbed to its highest level in nearly 30 years, traditional index crime rates experienced a historic drop.

Labor Force Analysis

Before diving into the Index Crime, first analyze the labor market as a major source of social pressure. We break this analysis down into three phases which are Pre-Pandemic (2016- 2019), Pandemic (2020-2021), and the Recovery Period (2022). The unemployment rate is more than just a statistic. It represents a break in the social contract and the accumulation of anxiety about survival. *Figure 6.1* below illustrates the one pages Labor Forces analysis dashboard.

Figure 6.1
Labor Force Dashboard



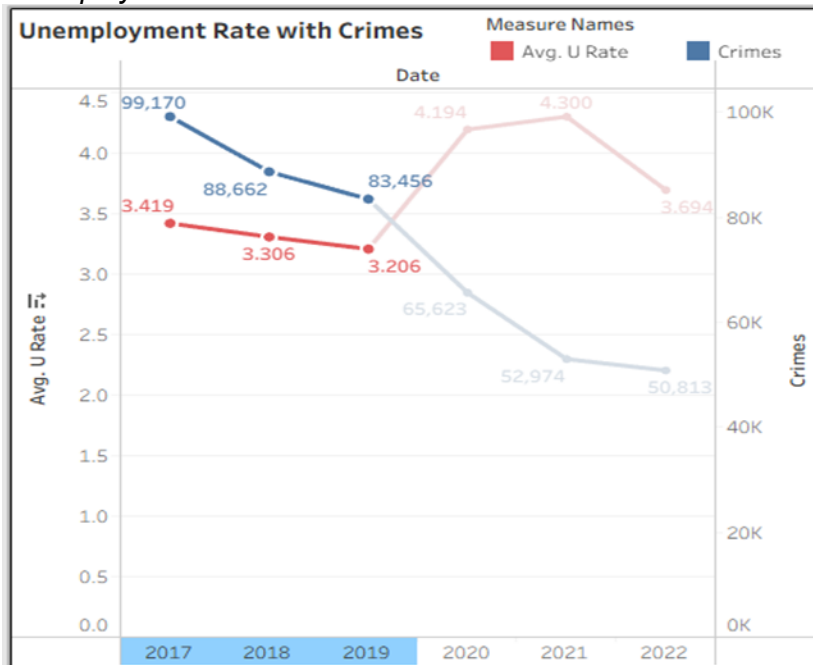
Note. The Labor Force Dashboard Analysis from 2017 to 2022, covering annual labor force data, the correlation between unemployment and crime rates (including Year-over-Year trends), state-level employment metrics, and a geographical map of unemployment distribution.

Pre-Pandemic (2016-2019)

From 2017 to 2019, Malaysia's macroeconomy remained relatively resilient. According to the Department of Statistics Malaysia (DOSM) (Dataset from Kaggle but reference to DOSM), the unemployment rate hovered between 3.234% (2019) and 3.4% (2017) illustrates as *figure 6.1* below. In economics, this is typically considered a level of 'full employment'.

Figure 6.1

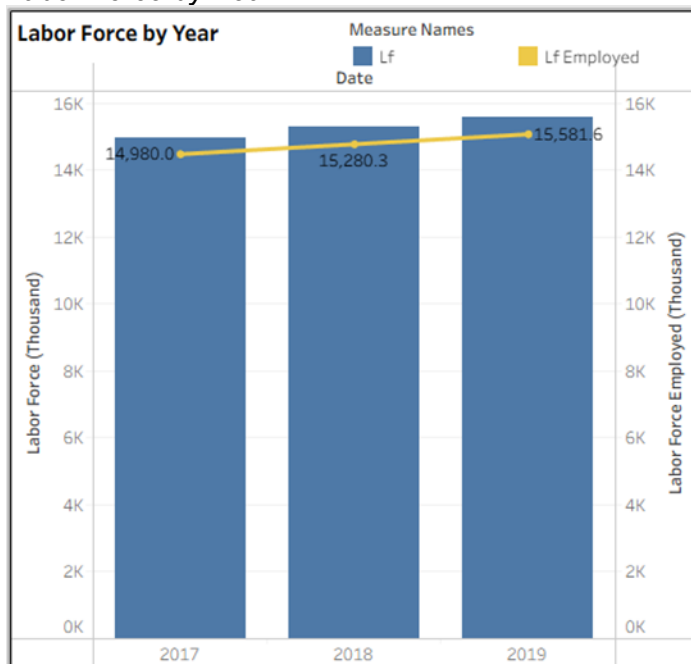
Unemployment Rate with Crimes



Note. This figure illustrates the relationship between the average unemployment rate (Avg. U Rate) and total crime cases from 2017 to 2019.

The labor force grew from 14,980 thousand people in 2017 (with a 3.419% unemployment rate) to a peak of 15,581.6 thousand people in 2019, as the unemployment rate improved to 3.206% illustrates as *figure 6.2* below. When drilling down to the state level, we observe that urban areas maintained higher employment rates compared to other regions. Notably, the top two highest average unemployment rates were recorded in East Malaysia Labuan 7.5% $((8.5 + 7.1 + 6.9) / 3)$ and Sabah 5.7% $((5.6 + 5.8 + 5.8) / 3)$ illustrates in *figure 6.3* below. According to the Sabah Employers Association, the lack of rail lines and quality roads hinder growth, with 50% business having five workers or less and leading to high unemployment (Free Malaysia Today, 2023).

Figure 6.2
Labor Force by Year



Note. This figure illustrates the annual growth of the total labor force (Lf) and the employed labor force (Lf Employed) from 2017 to 2019.

Figure 6.3
Unemployment Rate by State

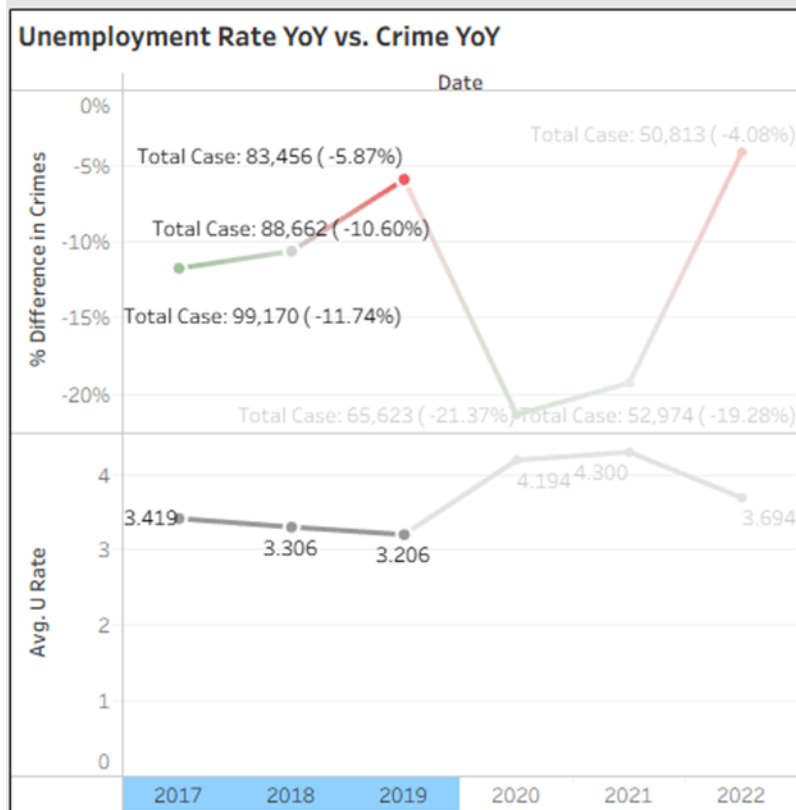


Note. This figure illustrates the annual growth of the total labor force (Lf) and the employed labor force (Lf Employed) from 2017 to 2019.

Overall, in year 2017-2019, Malaysia's unemployment rate showed a downward trend, which mirrors the movement of the crime rate displayed in *figure 6.1* (Unemployment Rate with Crime) above. The decline in crime appears to be directly proportional to the improvement in the labor market. As illustrated in the *figure 6.3* Unemployment Rate YoY vs. Crime YoY below, for every 0.1% decrease in the unemployment rate, the crime rate dropped by approximately 7%.

Figure 6.4

Unemployment Rate YoY vs. Crime YoY



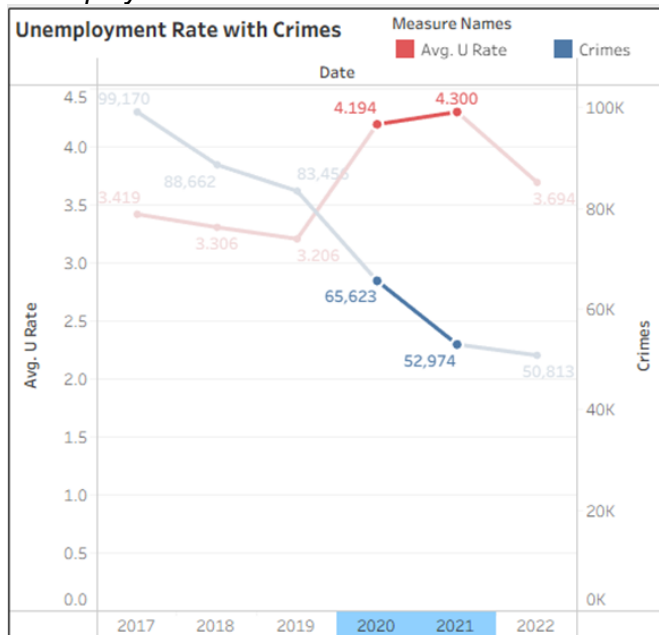
Note. This figure compares the year-over-year (YoY) percentage change in crime cases with the average unemployment rate (Avg. U Rate) from 2017 to 2019.

Pandemic (2020-2021)

In 2020, following the outbreak of the COVID-19 pandemic, the Prime Minister of Malaysia announced the implementation of the Movement Control Order (MCO) commencing on 18 March 2020. This period transitioned from a total economic lockdown to the gradual reopening of essential economic sectors, gradual reopening of key economic sectors, culminating in the lifting of interstate travel bans in October 2021, which marked the full resumption of domestic economic activities but the full resumption in April 2022.

Consequently, the unemployment rate surged by approximately 1%, reaching 4.19% in 2020 and peaking at 4.3% in 2021. Due to the strict movement restrictions imposed by the MCO, crime rates declined during this period. This specific phenomenon will be analyzed in greater detail in the upcoming 'Crime Analysis' section, as illustrated in the *figure 6.5* Unemployment Rate with Crime line chart below.

Figure 6.5
Unemployment Rate with Crimes



Note. This figure illustrates the average unemployment rate (Avg. U Rate) and total crime cases from 2020 to 2021

Furthermore, the Unemployment Rate by State table confirms that this was a universal trend, with every single state recording an increase in unemployment, as illustrated in the *figure 6.6* below.

Figure 6.6
Unemployment Rate by State

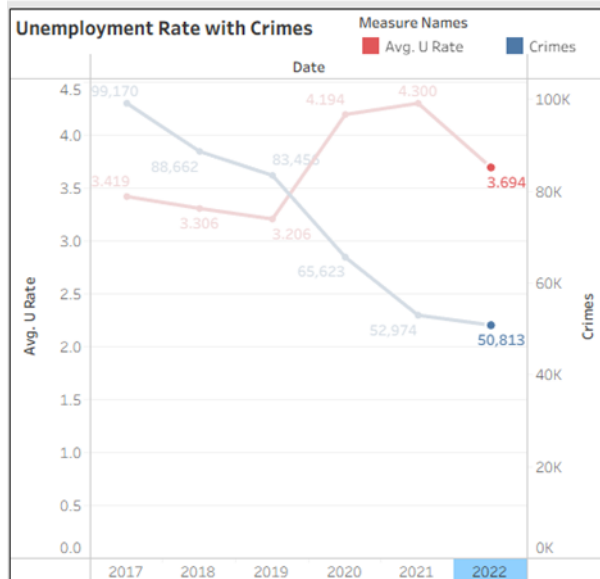


Note. This figure illustrates the unemployment rate (U Rate) across different states from 2019 to 2021, where red and orange highlights indicate higher rates.

Recovery Period (2022)

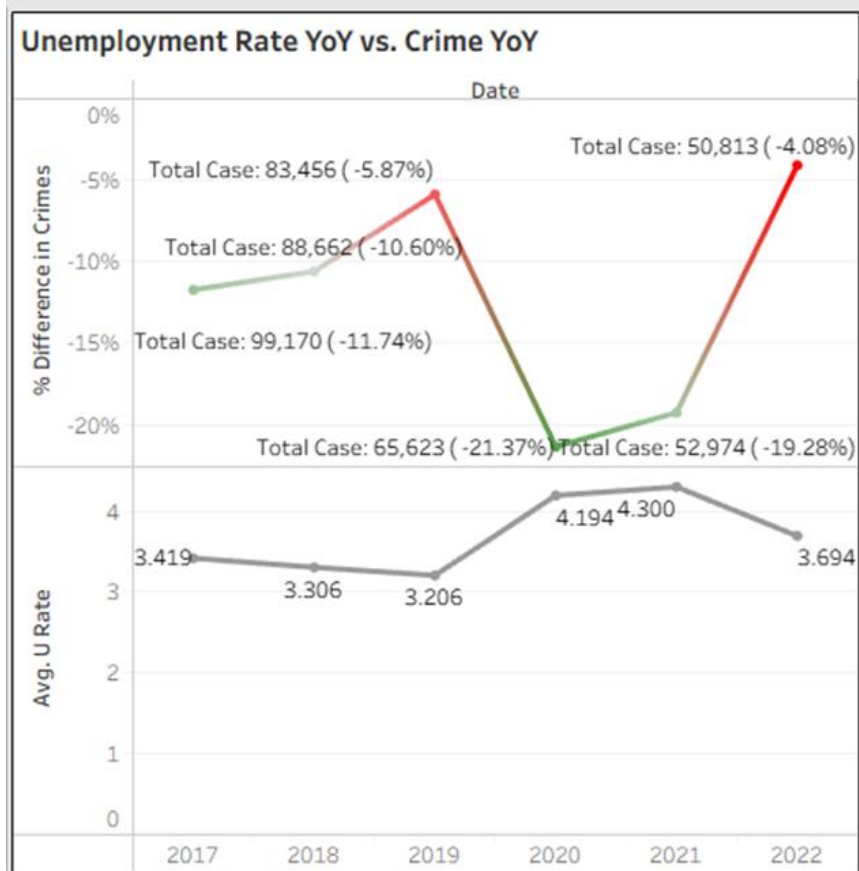
The reopening of borders in April 2022 and the full resumption of economic activities triggered a market recovery. The unemployment rate rapidly declined to 3.694% in 2022, effectively returning to pre-pandemic levels as illustrated in the *figure 6.7* below.

Figure 6.7
Unemployment Rate with Crimes



Note. This figure illustrates the average unemployment rate (Avg. U Rate) and total crime cases year 2022

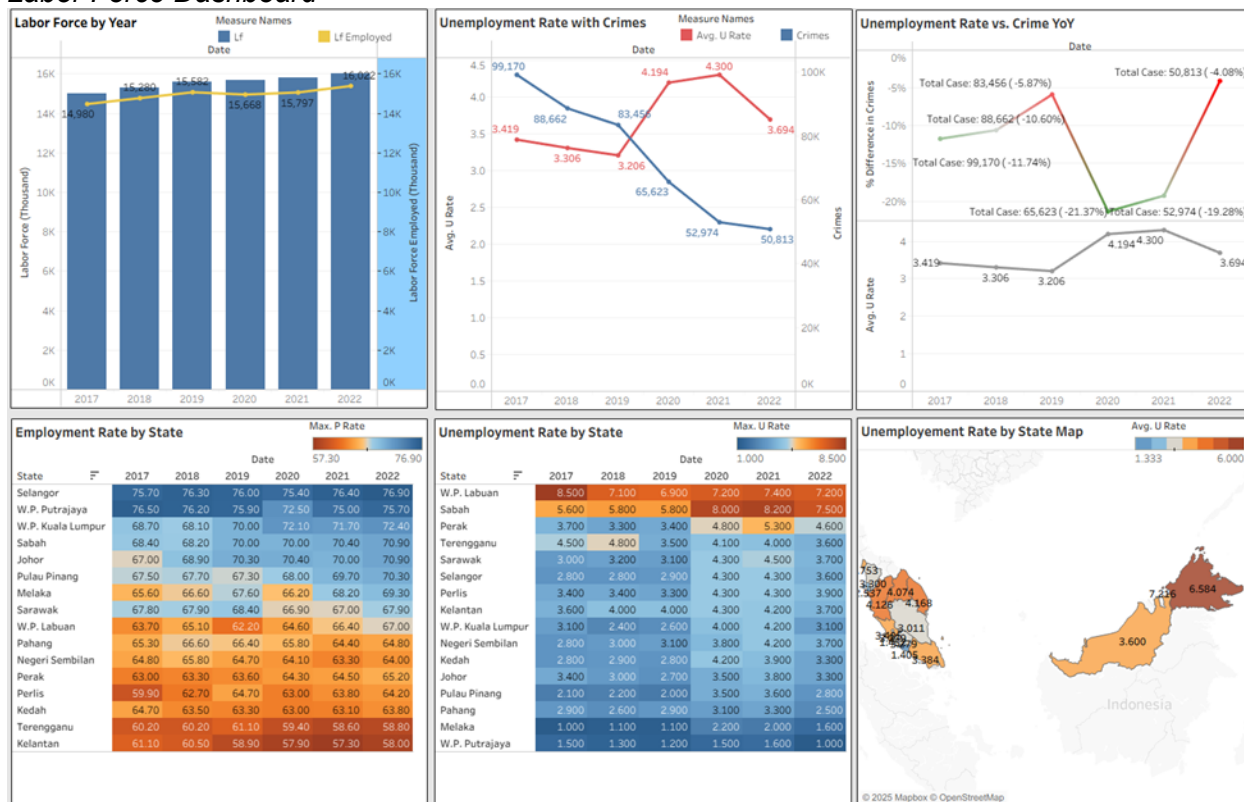
However, full employment in 2022 did not equate to financial well-being. Although people regained their jobs, they faced a drastically different cost-of-living environment and Inflationary pressures, particularly rising food and transport costs, eroded the purchasing power of wages (Department of Statistics Malaysia, n.d.-a). Regarding the data, although the crime rate decreased in 2022, the Unemployment Rate YoY vs. Crime YoY chart reveals a critical warning sign, the rate of decline in crime was the slowest (-4.08%) in five years as illustrated in the *figure 6.8* below. This deceleration was a precursor to a rebound. Indeed, our analysis of the Crime dataset confirms this trend, as crime rates recorded an increase in 2023.

Figure 6.8*Unemployment Rate YoY vs. Crime YoY*

Note. This figure compares the year-over-year (YoY) percentage change in crime cases with the average unemployment rate (Avg. U Rate) from 2017 to 2022.

Labour Dashboard Malaysia's labor force grew steadily from 14,980 thousand people in 2017 to 16,020 thousand people in 2019. The unemployment rate remained between 3.2% and 4.3%, peaking at 4.3% during the pandemic. When analysing the data at the state level, Labuan and Sabah recorded the highest unemployment rates. In Sabah, this is primarily due to underdeveloped environmental infrastructure, which discourages corporate investment. Conversely, Putrajaya recorded the lowest unemployment rate, largely due to its small population of approximately 109,202, compared to its neighbouring state's population of 6.99 million. Furthermore, there is a positive correlation between the unemployment rate and crime rates. As seen in the "Unemployment Rate vs. Crime YoY" chart, a significant drop in unemployment correlates with a larger decrease in the Crime YoY rate. During the pandemic, lockdown measures led to the largest crime reduction in five years at 21.37%. However, during the recovery period, the decrease in crime was at its lowest level in five years, which may be a precursor to a future increase in crime. *Figure 6.9* below illustrates the one pages Labor Forces analysis dashboard.

Figure 6.9
Labor Force Dashboard

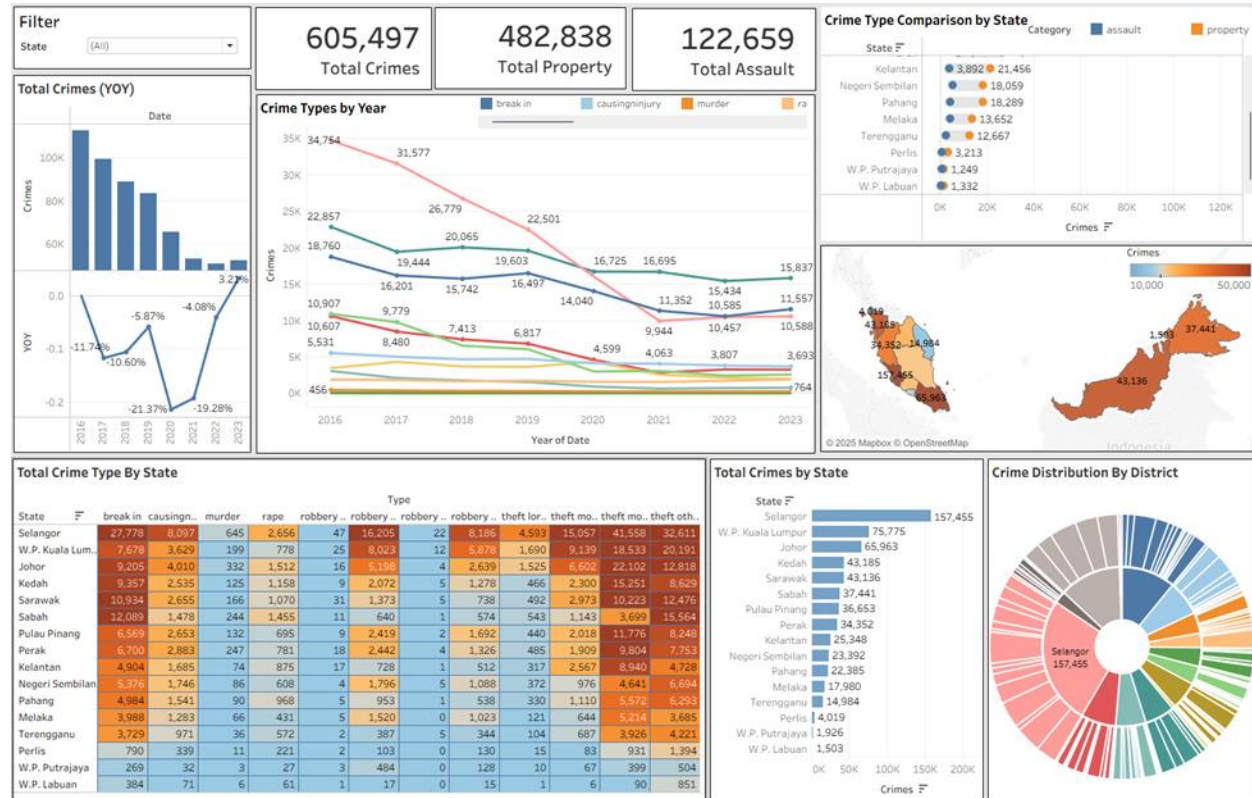


Note. The Labor Force Dashboard Analysis from 2017 to 2022, covering annual labor force data, the correlation between unemployment and crime rates (including Year-over-Year trends), state-level employment metrics, and a geographical map of unemployment distribution.

Crimes Analysis

In this section, we provide a comprehensive analysis of crime types and their distribution across Malaysia from 2016 to 2023. All analyses are integrated into a single Crimes Dashboard consisting of eight distinct visualizations, including YoY trends, state-level comparisons, and district-based radar charts to effectively evaluate the crimes that occurred during this period as illustrated in figure 6.10 below.

Figure 6.10
Crime Dashboard

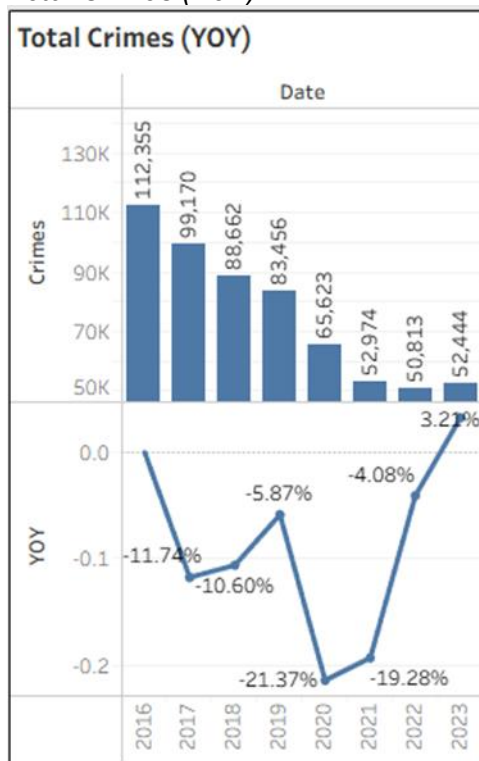


Note. The Crime Dashboard analysis from 2016 to 2023 covers total crime, crime types by year, crime type comparison by states, total crime type by state, total crimes by state, and crime distribution by district.

This section provides a multi-dimensional analysis of Malaysia's public safety status from 2016 to 2023, incorporating the latest data from the Crimes Dashboard.

As shown in Figure 6.11 below, the number of reported crime cases has demonstrated a year-on-year downward trend. During the COVID-19 pandemic, cases dropped significantly from 83,456 in 2019 to 65,623 in 2020, representing a decline of 21.37%.

Figure 6.11
Total Crimes (YoY)



Note. This figure illustrates the total volume of crime cases and the year-over-year (YoY) percentage change from 2016 to 2023.

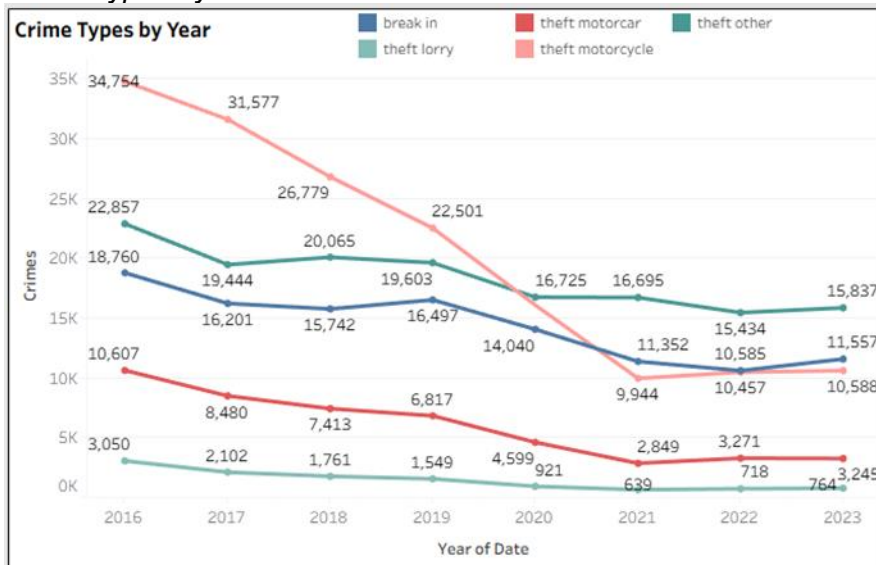
In 2021, the volume further decreased by 19.28% to 52,974 cases, largely attributed to the movement control measures. A total of 50,813 cases were recorded in 2022. Although Malaysia had fully reopened by this time, this year marked the smallest rate of decline over the seven-year period, a stabilization that often precedes a rebound. Subsequently, in 2023, crime cases rose by 3.2%, reaching 52,444.

From the year 2016 to 2023, total of 605,497 crime cases spanning the period from 2016 to 2023. Property Crime recorded a significant total of 482,838 cases, accounting for approximately 79.7%. This indicates that Malaysia's public safety challenges are primarily concentrated in economic-driven crimes, such as theft and burglary. The Property Crime category encompasses break in, theft of lorries, theft of motorcars, theft of motorcycles, and other thefts. Assault Crime comprised 122,659 cases, representing about 20.3%. Although this proportion is smaller, its impact on the public's sense of safety and psychological security is far more profound. This category includes causing injury, murder, rape, gang robbery with armed, solo robbery with armed, and solo robbery unarmed.

Property Crime

Within the property crime category, the Crime Types by Year line chart as illustrates in Figure 6.12 reveals that motorcycle theft was historically the most prevalent offense. In 2016, it peaked with 34,754 reported cases in 2016-2023. However, during the COVID-19 pandemic, this figure plummeted by approximately 71%, dropping to just 9,944 cases in 2021. Notably, in that same year, house break-ins surpassed motorcycle theft, recording 11,352 cases. The historical prevalence of motorcycle theft is largely attributed to the low barrier to entry and high market demand for spare parts, making it an attractive option for offenders (Budget Direct Insurance, n.d.). However, the pandemic drastically altered this landscape. The pandemic drastically disrupted the illicit supply chain. As workshops and parts dealers were closed under non-essential service rules, the black-market demand for stolen spare parts collapsed, directly causing a 71% drop in motorcycle thefts. Furthermore, strict movement controls and heightened police presence such as roadblocks and frequent patrols significantly increased the risk of detection for offenders, leading to the sharp decline observed.

Figure 6.12
Crime Types by Year



Note. This figure illustrates the total volume of property crime types from 2016 to 2023.

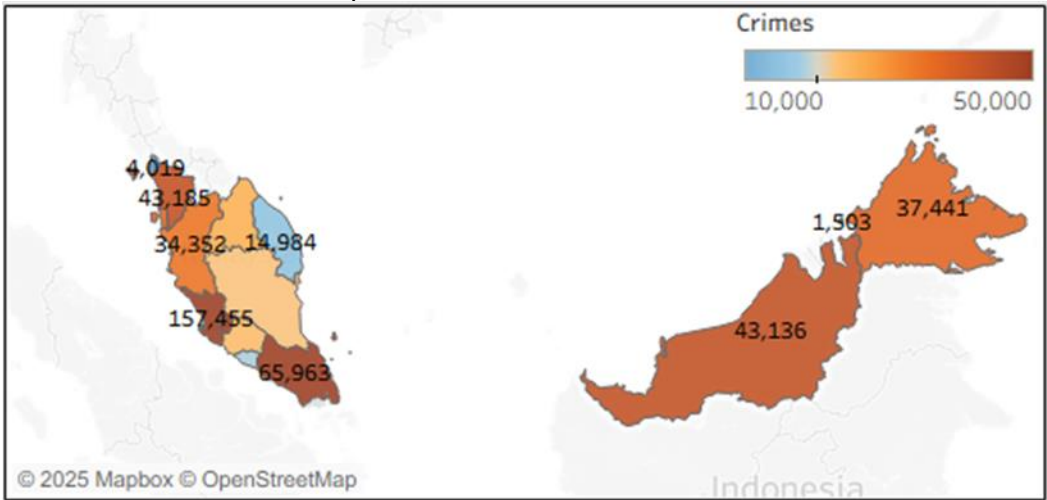
A state-level analysis of property crime from 2016 to 2023 reveals a distinct spatial disparity. Selangor consistently recorded the highest volume of cases across all categories including break-ins, vehicle thefts lorry, motorcar, motorcycle, and other thefts, far surpassing other states. Furthermore, the dense road networks inherent to such urbanized areas likely facilitate property crime by offering offenders convenient access and escape routes (Cohen & Felson, 1979). The high volume of property offenses in Selangor observed in this study (2016–2023) confirms that these spatial factors continue to drive crime concentration in the state. The results are illustrating in figure 6.13 Total Crime Type by State in table, figure 6.14 Distribution Map and figure 6.15 Total Crime Type by State in bar chart below.

Figure 6.13
Total Crime Type by State

Total Crime Type By State						
State	🇲🇾	Type				
		break in	theft lorry	theft motorcar	theft motorcycle	theft other
Selangor		27,778	4,593	15,057	41,558	32,611
W.P. Kuala Lum...		7,678	1,690	9,139	18,533	20,191
Johor		9,205	1,525	6,602	22,102	12,818
Sarawak		10,934	492	2,973	10,223	12,476
Kedah		9,357	466	2,300	15,251	8,629
Sabah		12,089	543	1,143	3,699	15,564
Pulau Pinang		6,569	440	2,018	11,776	8,248
Perak		6,700	485	1,909	9,804	7,753
Kelantan		4,904	317	2,567	8,940	4,728
Pahang		4,984	330	1,110	5,572	6,293
Negeri Sembilan		5,376	372	976	4,641	6,694
Melaka		3,988	121	644	5,214	3,685
Terengganu		3,729	104	687	3,926	4,221
Perlis		790	15	83	931	1,394
W.P. Labuan		384	1	6	90	851
W.P. Putrajaya		269	10	67	399	504

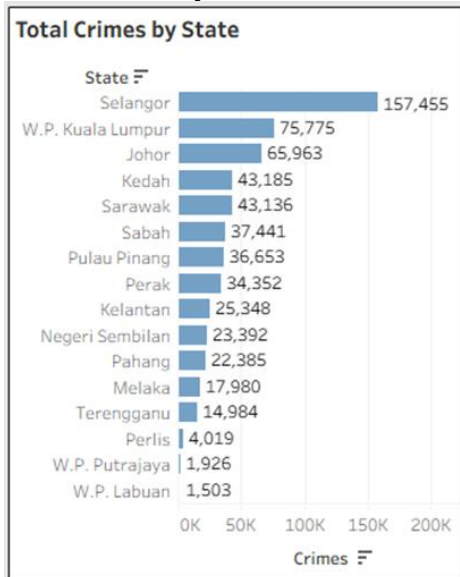
Note. This figure illustrates the breakdown of total property crime types across different states from 2016 to 2023, where darker red highlights indicate higher numbers of reported cases.

Figure 6.14
Total Crime distribution map



Note. This figure illustrates the geographical distribution of total reported property crime cases across Malaysia from 2016 to 2023, where darker red regions represent states with higher crime volumes.

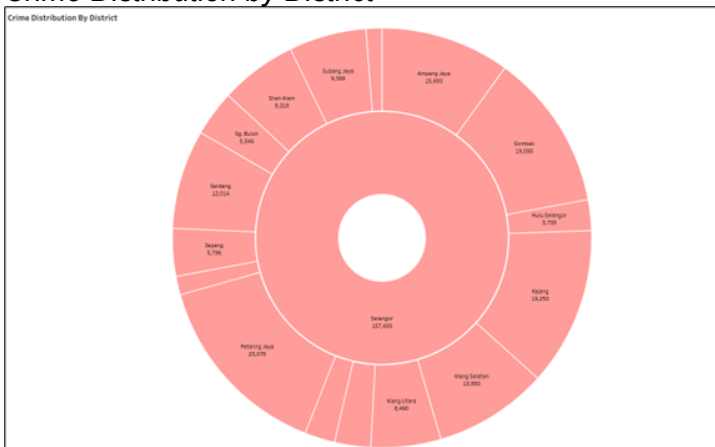
Figure 6.15
Total Crimes by State



Note. This figure illustrates the total volume of reported property crime cases by state from 2016 to 2023, ranked in descending order.

By analysing the Crime Distribution by District in *figure 6.16*, we can drill down into the Selangor state level to identify specific high-risk zones. The data reveals that Petaling District ranks first with the highest recorded crime volume. Petaling is the most populous district, housing approximately 3.6 million residents with a high population. The district features an intricate and complex road network, serving as a convergence point for major highways such as the Federal Highway, Sistem Penyuraian Trafik KL Barat (Sprint), New Pantai Expressway, and Lebuhraya Damansara-Shah Alam (DASH). As highlighted in (Cohen & Felson, 1979), the combination of high population density providing abundant targets and complex urban infrastructure facilitating offender mobility effectively transforms Petaling into the primary hotspot for criminal activity in the state

Figure 6.16
Crime Distribution by District

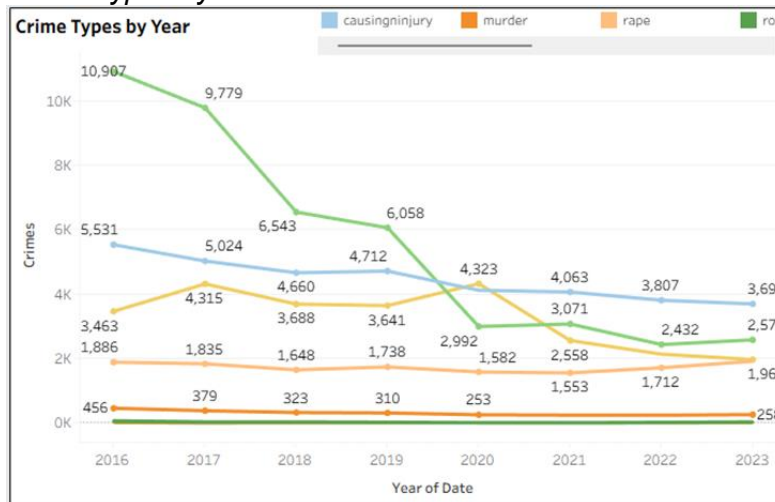


Note. This figure illustrates the hierarchical distribution of reported crime cases across various districts within the state of Selangor, highlighting specific high-risk zones.

Assault Crime

On the other hand, an analysis of Assault Crime includes causing injury, murder, rape, gang robbery with armed, gang robbery unarmed, solo robbery with armed, and solo robbery unarmed. Among these seven categories, unarmed gang robbery was historically the highest, reaching 10,907 cases in 2016. Before the pandemic, causing injury was the second highest with 5,531 cases. However, a strange phenomenon occurred during the COVID-19 pandemic. While the other six crime types decreased, solo robbery unarmed actually increased by 30%. In 2020, it became the most frequent assault crime, with a total of 4,323 cases.

Figure 6.17
Crime Types by Year



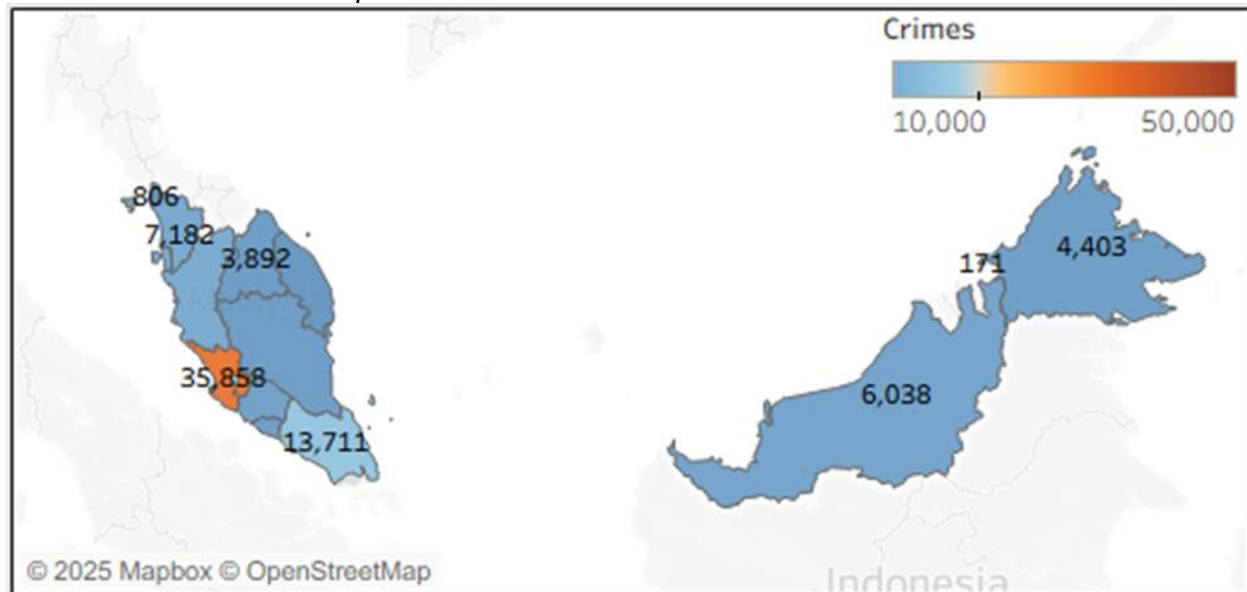
Note. This figure illustrates the total volume of assault crime types from 2016 to 2023.

Drilling down to the state level, as presented in *figure 6.18* Total Crime Type By State and the *figure 6.19* Distribution Map, Selangor clearly emerges as the epicenter of criminal activity. It records the highest number of cases, with a particularly high incidence of unarmed gang robbery.

Figure 6.18
Total Crime Type by State

State	Type						
	causinginjury	murder	rape	robbery gang a..	robbery gang u..	robbery solo ar..	robbery solo un..
Selangor	8,097	645	2,656	47	16,205	22	8,186
W.P. Kuala Lum..	3,629	199	778	25	8,023	12	5,878
Johor	4,010	332	1,512	16	5,198	4	2,639
Perak	2,883	247	781	18	2,442	4	1,326
Pulau Pinang	2,653	132	695	9	2,419	2	1,692
Kedah	2,535	125	1,158	9	2,072	5	1,278
Sarawak	2,655	166	1,070	31	1,373	5	738
Negeri Sembilan	1,746	86	608	4	1,796	5	1,088
Sabah	1,478	244	1,455	11	640	1	574
Melaka	1,283	66	431	5	1,520	0	1,023
Pahang	1,541	90	968	5	953	1	538
Kelantan	1,685	74	875	17	728	1	512
Terengganu	971	36	572	2	387	5	344
Perlis	339	11	221	2	103	0	130
W.P. Putrajaya	32	3	27	3	484	0	128
W.P. Labuan	71	6	61	1	17	0	15

Note. This figure illustrates the breakdown of total assault crime types across different states from 2016 to 2023, where darker red highlights indicate higher numbers of reported cases.

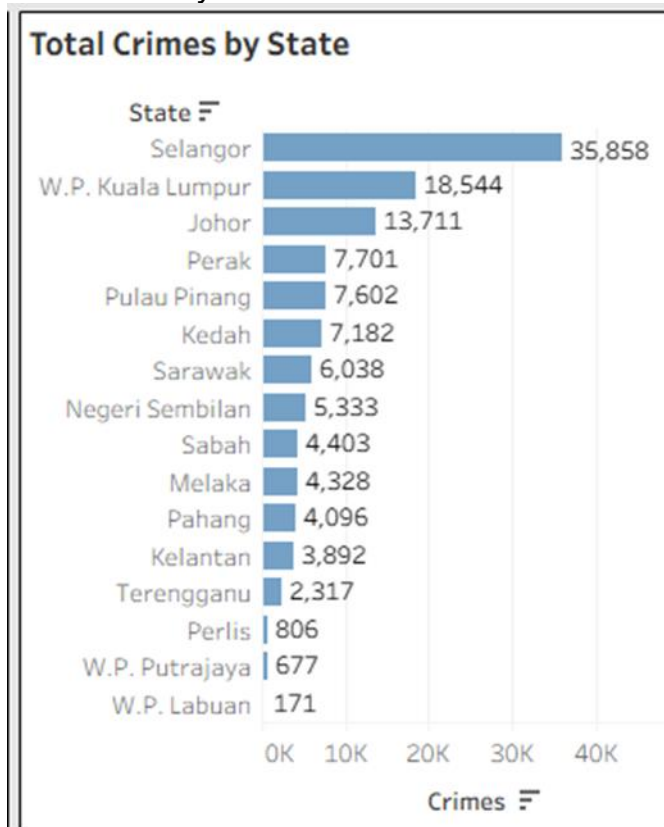
Figure 6.19*Total Crime distribution map*

Note. This figure illustrates the geographical distribution of total reported assault crime cases across Malaysia from 2016 to 2023, where darker red regions represent states with higher crime volumes.

The bar chart illustrates in *figure 6.20* below highlights a stark disparity in crime distribution across Malaysia. A clear polarization is evident while the top three states led by Selangor and Kuala Lumpur record exponentially higher crime figures, the remaining states maintain a comparatively low profile, with total cases consistently falling below the 10,000 marks over the 8-year period.

As noted in previous studies above, unemployment, economic distress in highly urbanized environments often acts as a catalyst for property crime, as marginalized individuals resort to illicit means for survival. Consequently, these factors collectively entrench Selangor and Kuala Lumpur as the primary hotspots for criminal activity in the country.

Figure 6.20
Total Crimes by State



Note. This figure illustrates the total volume of reported assault crime cases by state from 2016 to 2023, ranked in descending order.

In summary, by synthesizing the Labor Force and Crime Analysis, the data can be categorized into three distinct phases the Pre-Pandemic, Pandemic, and Recovery periods.

During the Pre-Pandemic phase, unemployment was the primary driver of crime. However, during the Recovery phase 2022, despite low unemployment, inflationary pressures and rising living costs became the key catalysts preventing a further decline in crime rates. The Pandemic period 2020–2021 presented a notable anomaly, while overall crime rates declined due to MCO restrictions, it is crucial to note that specific desperation-driven crimes, such as unarmed solo robbery, surged. This highlights that economic strain still drove criminal behavior, even during lockdown.

From a state-level perspective, Selangor and Kuala Lumpur as the most densely populated regions in Malaysia, face the unique challenges of rapid urbanization. The high concentration of population in these economic hubs inevitably exacerbates issues related to unemployment and socio-economic disparity. Consequently, these structural factors explain why Selangor and Kuala Lumpur consistently record significantly higher crime rates compared to other states.

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Appendices

Final Visualization Dashboard URL:

https://public.tableau.com/app/profile/chong.keat.lee/viz/USM_505_Project_final/Story1